

## Batch Normalization

Batch Normalization is used to reduce the problem of **internal covariate shift** in neural networks. It works by normalizing the data within each mini-batch. This means it calculates the mean and variance of data in a batch and then adjusts the values so that they have similar range. After that it scales and shifts the values so that model learn effectively.

In traditional neural networks as the input data propagates through the network, the distribution of each layer's inputs changes. This phenomenon is known as internal covariate shift and it can slow down training process. Batch Normalization aims to reduce this issue by normalizing the inputs of each layer.

This process keeps the inputs to each layer of the network in a stable range even if the outputs of earlier layers change during training. As a result training becomes faster and more stable.

## Need of Batch Normalization

Batch Normalization makes sure outputs of each layer stay steady as model learns. This helps model train faster and learn more effectively.

- Solves the problem of internal covariate shift.
- Makes training faster and more stable.
- Allows use of higher learning rates.
- Helps avoid vanishing or exploding gradients.
- Can act like a regularizer sometimes reduce the need for dropout.

## Fundamentals of Batch Normalization

In this section we are going to discuss the steps taken to perform batch normalization.

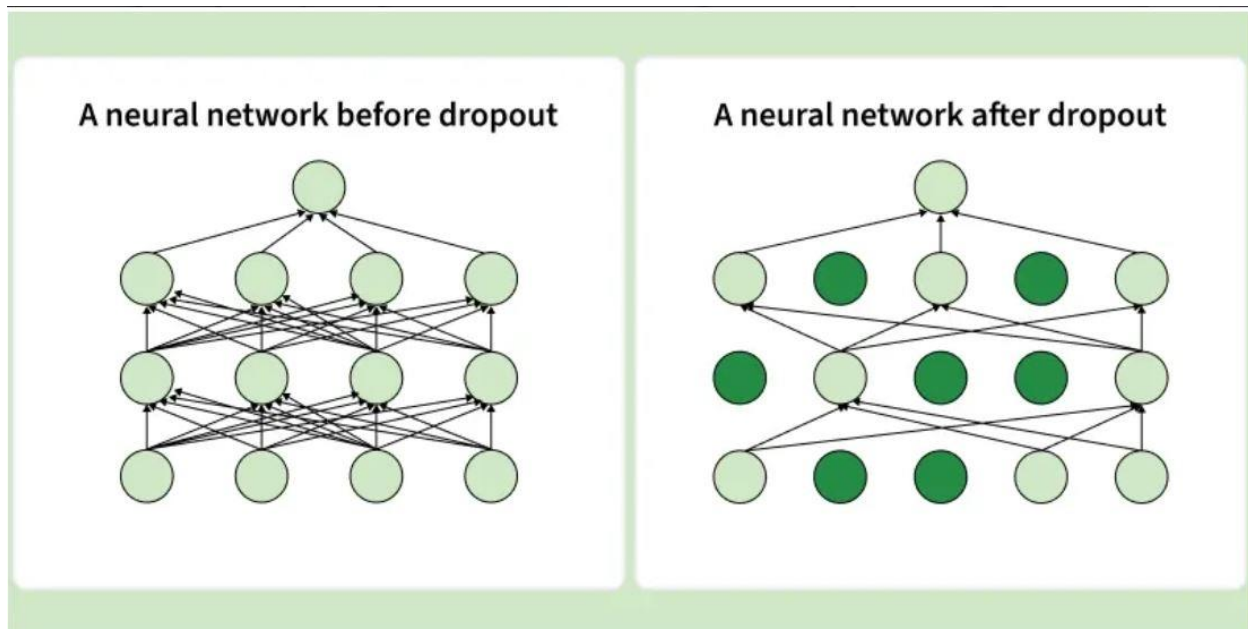
Step 1: Compute the Mean and Variance of Mini-Batches

Step 2: Normalization

Step 3: Scale and Shift the Normalized Activations

## Dropout

Training deep learning models for too long on the same data can lead to overfitting where the model performs well on training data but poorly on unseen data. Dropout Regularization helps overcome this by randomly deactivating a portion of neurons during training hence forcing the model to learn more robust and independent features.



## Why Dropout is needed

Dropout is needed because it prevents the network from relying too much on specific neurons, hence improving generalization by training and reducing co-adaptation among neurons. Below are the key features of dropout:

- Randomly deactivates a set percentage of neurons in each training step.
- Works with dense, convolutional and recurrent layers.
- Dropout rate controls how many neurons are dropped.
- Remaining active neurons are scaled to maintain output stability.
- Reduces co-dependency among neurons and improves generalization.

## Types of Dropout Regularization

- **Standard Dropout:** Randomly removes individual neurons during training to reduce overfitting.
- **Spatial Dropout:** Drops entire feature maps in CNNs to preserve spatial structure.

- **Dropout2D / Dropout3D:** Applies dropout across 2D or 3D feature channels for images and volumetric data.
- **Drop Connect:** Randomly drops weights instead of neurons, offering stronger regularization.
- **Alpha Dropout:** Maintains activation statistics and works best with SELU-based networks.
- **Variational Dropout:** Uses the same dropout mask across time steps in RNNs for stable learning.
- **Monte Carlo Dropout:** Keeps dropout active at inference to estimate prediction uncertainty.